

7. The finiteness theorem for the invariants of finite groups

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In what follows an entirely elementary proof of finiteness for the invariants of finite groups is to be given — one based only on the theory of symmetric functions — and at the same time it gives an actual specification of the full system; whereas the usual proof, based on Hilbert’s theorem on module bases (Ann. 36), is only an existence proof.¹

Let the finite group $\mathfrak{H}$ consist of the h linear transformations (with non-vanishing determinant) $A_1, \dots, A_h$, where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^n a_{1\nu}^{(k)} x_\nu, \dots, x_n^{(k)} = \sum_{\nu=1}^n a_{n\nu}^{(k)} x_\nu,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. The group $\mathfrak{H}$ therefore carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must be contained among $A_1, \dots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$. By an integral rational (absolute) invariant of the group is meant an integral rational function of $x_1, \dots, x_n$ which remains identically unchanged under the application of $A_1, \dots, A_h$; for such an invariant $f(x)$ one therefore has

$$f(x) = f(x^{(1)}) = \dots = f(x^{(h)}) = \frac{1}{h} \sum_{k=1}^h f(x^{(k)}). \quad (1)$$

1.

Formula (1) says that $f(x)$ is an integral rational symmetric function of the rows of quantities $(x^{(1)}), \dots, (x^{(h)})$ — and indeed, since each summand $f(x^{(k)})$ contains only the one row $(x^{(k)})$, this is the simplest case, usually called the *one-form* case. By the known theorem on symmetric functions of rows of quantities,² $f(x)$ is therefore integrally and rationally representable by the elementary symmetric functions of these rows, that is, by the coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the “Galois resolvent”

$$\begin{aligned} \Phi(z, u) &= \prod_{k=1}^h (z + u_1 x_1^{(k)} + \dots + u_n x_n^{(k)}) \\ &= z^h + \sum G_{\alpha\alpha_1\dots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \dots u_n^{\alpha_n}, \end{aligned} \quad \left(\begin{array}{c} \alpha + \alpha_1 + \dots + \alpha_n = h \\ \alpha \neq h \end{array} \right),$$

where the $G_{\alpha\alpha_1\dots\alpha_n}(x)$ are invariants of degree $\alpha_1 + \dots + \alpha_n$ in the x ’s. Thus it is proved:

The coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the Galois resolvent form a full system of invariants of the group, in such a way that every invariant can be represented integrally and rationally by these finitely many invariants.

2.

Starting from (1), one may also make the following still more elementary consideration,³ which at the same time leads to a second full system. Put

$$f(x) = a + bx_1^{\mu_1} \dots x_n^{\mu_n} + \dots + cx_1^{\nu_1} \dots x_n^{\nu_n},$$

where $a, b, \dots, c$ denote constants. Then by (1) one has

$$\begin{aligned} h \cdot f(x) &= h \cdot a + b \sum_{k=1}^h (x_1^{(k)})^{\mu_1} \dots (x_n^{(k)})^{\mu_n} + \dots + c \sum_{k=1}^h (x_1^{(k)})^{\nu_1} \dots (x_n^{(k)})^{\nu_n} \\ &= h \cdot a + b \cdot J_{\mu_1\dots\mu_n} + \dots + c \cdot J_{\nu_1\dots\nu_n}. \end{aligned}$$

Every invariant can therefore be composed integrally and linearly from the special invariants

$$J_{\mu_1\dots\mu_n} = \sum_{k=1}^h (x_1^{(k)})^{\mu_1} \dots (x_n^{(k)})^{\mu_n},$$

¹Cf., for example, Weber, *Lehrbuch der Algebra* (2nd ed.), vol. 2, § 57.

²Cf. the note to 2.

³This amounts to a proof of the theorem on symmetric functions of rows of quantities in the above-mentioned one-form case.

and it is enough to prove the assertion for these special ones. But, apart from a numerical factor, $J_{\mu_1 \dots \mu_n}$ is the coefficient of $u_1^{\mu_1} \dots u_n^{\mu_n}$ in the expression

$$S_\mu = \sum_{k=1}^h (u_1 x_1^{(k)} + \dots + u_n x_n^{(k)})^\mu,$$

where $\mu = \mu_1 + \dots + \mu_n$, and this expression is the μ -th power sum of the h linear forms

$$\xi_1 = u_1 x_1^{(1)} + \dots + u_n x_n^{(1)}, \dots, \xi_h = u_1 x_1^{(h)} + \dots + u_n x_n^{(h)}.$$

It is known that the infinitely many power sums S_μ are integral rational combinations of

$$S_1, S_2, \dots, S_h,$$

whose coefficients are given by the invariants

$$J_{\mu_1 \dots \mu_n} \quad (\mu_1 + \dots + \mu_n \leq h).$$

Thus a second full system has been obtained:

A full system of invariants of the group is given by all invariants $J_{\mu_1 \dots \mu_n}$ for which $\mu_1 + \dots + \mu_n \leq h$, where h denotes the order of the group.

The connection with 1. is established by observing that the power sums $S_1, \dots, S_h$ can be replaced by the elementary symmetric functions of $\xi_1, \dots, \xi_h$, which leads to the full system of the $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ given there. Both results show that all invariants can be expressed integrally and rationally by those whose degree in the x 's does not exceed the order h of the group.

3.

Consequences for rational representation can be drawn from what has just been proved. Every rational absolute invariant can be represented — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally by the coefficients $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$. This theorem can also be proved easily in various ways without passing through the finiteness theorem; it is already formulated in Weber II, § 58.⁴

4.

Finally, it should be mentioned that the results derived here are implicitly contained in my paper “Fields and systems of rational functions”;⁵ more precisely, the full system given in 1. is contained in Theorems VI and VII, and a proof of rational representation independent of this finiteness theorem is contained in Theorem III.⁶ In the present special case, however, the course of the proof and the result simplify considerably in comparison with the general theory. I was led to apply the general investigations to the invariants of finite groups by conversations with E. Fischer, from whom, in substance, the proof given under 2 — in the general case the full system that appears there is not rationally definable — and the note to 3 also originate.

Erlangen, May 1915.

⁴The proof given there, however, is not sound. It is only shown that the function $\Psi(t)$ in formula (7) has invariants as coefficients, but not that these invariants are rationally expressible by the coefficients of $\Phi(t)$. This gap is avoided if, in representing the $x_i^{(k)}$ by ξ_k , one uses a known differentiation procedure instead of Lagrange's interpolation formula. This is the relation obtained by differentiating the identity $\Phi(-\xi_k, u) = 0$ with respect to all u :

$$\left[\frac{\partial \Phi}{\partial u_i} - x_i^{(k)} \frac{\partial \Phi}{\partial z} \right]_{z=-\xi_k} = 0.$$

Accordingly, for each rational function $\omega(x)$ the following formula must replace formulas (7) and (8) in Weber:

$$\omega(x_1^{(k)}, \dots, x_n^{(k)}) = \omega \left(\frac{\partial \Phi / \partial u_1}{\partial \Phi / \partial z}, \dots, \frac{\partial \Phi / \partial u_n}{\partial \Phi / \partial z} \right) \bigg|_{z=-\xi_k},$$

which contains only coefficients of $\Phi(z, u)$, and whose summation over all k , for invariants $\omega(x)$, gives the desired representation.

⁵Math. Ann. 76, p. 161 (1915).

⁶For relative invariants, Theorems VIII and IX contain a proof of finiteness which is likewise based on a different foundation from the usual one, but is also only an existence proof; the reason is that the relative invariants do not form a field.

8. On integral rational representation of the invariants of a system of arbitrarily many ground forms

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At the end of his contribution to the Schwarz Festschrift, “On the invariants of a system of arbitrarily many ground forms”, D. Hilbert states the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ) , where J denotes the full system of invariants of the linearly independent ground forms, and the invariants PJ arise from J by polar processes.

I show below, in I, that this conjecture is in fact correct, and indeed as a simple consequence of the reduction theorem that every form in arbitrarily many rows of n variables each can be represented as a sum of polars of forms which contain only n fixed rows and which themselves are derived from the original form by polar processes. This reduction theorem is a special case, first formulated by F. Mertens, of the generalization, given by A. Capelli, F. Mertens, and J. Deruyts, of the Clebsch–Gordan series expansion to forms in arbitrarily many rows of n variables each; this generalization is proved by formal transformation of the differentiation processes.⁷

In II I also give a more conceptual proof of the reduction theorem, by treating the question as a problem concerning the equivalence of linear families of forms. This point of view, and also an essential part of the considerations used, I take from a paper “On algebraic module systems”⁸ and from unpublished theorems of E. Fischer adjoining it, whose use he kindly permitted me.

Finally, in III, I show how the corresponding considerations also lead to the most general series expansions.

I.

Let θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \dots, A_N,$$

where in general the row A_k consists of the n elements

$$A_k^{(1)}, A_k^{(2)}, \dots, A_k^{(n)}.$$

The coefficients of θ may be indeterminates or quantities of a given domain of rationality. Let P further denote an integral rational function — with rational integer coefficients — of the polar processes

$$P_{hk} = \left(A_h \frac{\partial}{\partial A_k} \right) = A_h^{(1)} \frac{\partial}{\partial A_k^{(1)}} + A_h^{(2)} \frac{\partial}{\partial A_k^{(2)}} + \dots + A_h^{(n)} \frac{\partial}{\partial A_k^{(n)}},$$

where the product $P_{hk}P_{ij}\theta$ is to be understood as the application of P_{hk} to $P_{ij}\theta$.

Then the reduction theorem just mentioned is given by the identity

$$\theta = \sum PZ, \tag{1}$$

where the forms Z contain only the rows

$$A_1, A_2, \dots, A_n$$

and are derived from θ by polar processes P .

We now consider the system of ground forms (F') , consisting of the N forms of the same order and the same number of variables

$$F_1, F_2, \dots, F_N,$$

whose coefficients are taken respectively as the N rows

$$A_1, A_2, \dots, A_N.$$

⁷F. Mertens: “Über eine Formel der Determinantentheorie.” (Formula 5), Sitzb. d. Ak. d. Wiss. Wien, vol. 91, II. Abt. (1885), and more fully (with applications) in: “Über ganze Funktionen von m Systemen von je n Unbestimmten”. Monatsh. f. Math. u. Phys. 4th year (1893). For some of Capelli’s proofs (since 1882) cf., for example, Math. Ann. 37, or the autographed “Lezioni sulla teoria delle forme algebriche” (1902). J. Deruyts: *Essai d’une théorie générale des formes algébriques*, Mém. d. 1. Soc. Roy. des Sciences de Liège (2) 17 (1892).

⁸E. Fischer: Über algebraische Modulsysteme und lineare homogene partielle Differentialgleichungen mit konstanten Koeffizienten, § 1–4, J. f. M. 140 (1911).

Let (J) denote the full system — consisting of finitely many invariants — of $F_1, \dots, F_n$, in such a way that every invariant of $F_1, \dots, F_n$ can be expressed integrally and rationally by the invariants (J) . Hilbert's conjecture to be proved is then that, for the simultaneous invariants S of $F_1, \dots, F_N$, a full system is given by the finitely many invariants (J, PJ) , where PJ denotes all invariants derivable from J by polar processes P .

Indeed, every simultaneous invariant S with rational integer coefficients of (F') is a form homogeneous in each of the N rows $A_1, \dots, A_N$ — with rational integer coefficients — and identity (1) can therefore be applied to it. Since application of the polar processes P to S is known to produce invariants again, the forms Z are also invariants, namely simultaneous invariants of $F_1, \dots, F_n$, since they contain only the rows $A_1, \dots, A_n$; hence, by assumption, they are integral rational functions of the invariants (J) . Thus identity (1) becomes

$$S = \sum PY,$$

where the Y denote power products of the invariants (J) . But, by the definition of P , the actual execution of the processes P on Y consists in the successive, finitely repeated execution of the simple polar operations P_{hk} , which, by the differentiation rule for products, leads to sums of products of invariants (J) and (PJ) . The same holds for the application of P_{hk} to a product from (J) and (PJ) ; hence PY too yields only integral rational combinations of (J, PJ) . *The integral rational representability of all invariants S by (J, PJ) is thereby proved.*

II.

Before proving the reduction theorem, we recall some facts about linear families of forms. By a *linear family of forms over K* we mean a system of forms of the same dimension, complete in the sense that together with θ_1 and θ_2 it always contains $c_1\theta_1 + c_2\theta_2$, where c_1, c_2 are quantities of a prescribed domain of rationality K , and where K contains the coefficient domain of the θ 's. Among these forms there is always a finite number — say ρ — linearly independent over K , while any $\rho + 1$ forms satisfy a linear relation with coefficients from K ; the family has *rank* ρ with respect to K .⁹ We shall use the easily seen fact that if $\mathcal{L}$ and $\mathcal{T}$ are two linear families over K of the same rank ρ , and if $\mathcal{T}$ is a subfamily of $\mathcal{L}$, then $\mathcal{L}$ and $\mathcal{T}$ are identical (equivalent).

With this in hand, we pass to the reduction theorem. The identity from I,

$$\theta = \sum PZ,$$

passes into an analogous identity for $P\theta$ when one applies the same polar process P to both sides; thus the reduction theorem at the same time gives a representation of all polars of θ by sums of the special polars PZ . Since, conversely, these special polars are surely contained among all polars, the reduction theorem asserts the equivalence of these two linear families of forms; in particular, it also asserts the equivalence of those subfamilies whose forms have, in each individual row, the same dimension as θ . To prove this equivalence we insert an intermediate family determined by the forms Z . For simplicity we first give the proof for the case of three binary rows ($N = 3, n = 2$), and then briefly indicate the completely parallel general proof.

Thus let $\theta(x, y, z)$ have dimensions α, β, p respectively in the rows

$$x_1x_2; \quad y_1y_2; \quad z_1z_2;$$

and let the coefficients of θ first be indeterminates a (or rational numbers). We consider the following three linear families of forms.

1. The family $\mathcal{L}$, consisting of all forms $f(x, y, z)$ which arise from θ by polar processes P and which, in the individual rows x, y, z , have the same dimension as θ ; $\mathcal{L}$ contains θ in particular. If $f(x, y, z)$ is regarded as a form in x, y, z and the indeterminates a , then the coefficient field is the field R of rational numbers; for K as well one must take R , since only for rational integral θ_1, θ_2 does $c_1P_1 + c_2P_2$ again become a process P ; let $\mathcal{L}$ have rank ρ with respect to R .

2. The family $\mathcal{S}$, consisting of all forms $g(\lambda; x, y)$ defined by

$$g(\lambda; x, y) = f(x, y, \lambda_1x + \lambda_2y),$$

or, expressed by polar processes,

$$\begin{aligned} g(\lambda; x, y) &= \frac{1}{p!} \left[(\lambda_1x + \lambda_2y) \frac{\partial}{\partial z} \right]^p f(x, y, z) \\ &= g_0(xy)\lambda_1^p + g_1(x, y)\lambda_1^{p-1}\lambda_2 + \dots + g_p(xy)\lambda_2^p, \end{aligned}$$

⁹More general linear families of forms — whose rank need not be finite — are obtained by dropping the restriction of equal dimension, or also by dropping the restriction that K contain the coefficient domain of the θ 's.

where

$$i!(p-i)!g_i(xy) = \left(y \frac{\partial}{\partial z}\right)^i \left(x \frac{\partial}{\partial z}\right)^{p-i} f(xyz).$$

¹⁰ Thus the $g_i(xy)$ give forms Z of identity (1). The g are rational-integer forms in x, y, λ and the indeterminates a ; let $\mathcal{S}$ have rank σ with respect to R .

3. The family $\mathcal{T}$, obtained from $\mathcal{S}$ by the inverse polar process just as $\mathcal{S}$ is obtained from $\mathcal{L}$; the forms h of $\mathcal{T}$ are defined by

$$\begin{aligned} h(x, y, z) &= \frac{1}{p!} \left[z \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y} \right) \right]^p g(\lambda; xy) \\ &= h_0(xyz) + h_1(xyz) + \cdots + h_p(xyz), \end{aligned}$$

where, by 2.,

$$h_i(xyz) = \left(z \frac{\partial}{\partial x}\right)^{p-i} \left(z \frac{\partial}{\partial y}\right)^i g_i(xy).$$

The individual h_i , and consequently also h , are therefore forms PZ and sums of them; let $\mathcal{T}$ have rank τ with respect to R .

As the construction of the forms $h(x, y, z)$ of $\mathcal{T}$ shows, these forms arise from θ by polar processes P , and in the rows x, y, z have individually the same dimension as θ ; hence the family $\mathcal{T}$ is a subfamily of $\mathcal{L}$. By the fact mentioned above, it remains only to prove equality of the ranks. In this proof no further use is made of the special structure of the $f(xyz)$ — namely, that they are polars of one and the same form θ .

a) For comparing the ranks of $\mathcal{L}$ and $\mathcal{S}$ we use Lemma a): from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ it follows necessarily that $f(x, y, z) = 0$. This follows immediately from the identical relation between three binary rows

$$(xy)z + (yz)x + (zx)y = 0, \quad (xy) = (x_1 y_2 - x_2 y_1), \dots,$$

which implies

$$f[x, y, (xy)x + (xz)y] = (xy)^p f(x, y, z).$$

Thus from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ (identically in λ) it follows, as is shown by the substitution $\lambda_1 = (zy)$, $\lambda_2 = (xz)$, and since $(xy) \neq 0$, that necessarily $f(x, y, z) = 0$.

By this lemma there is a one-to-one correspondence between the forms of $\mathcal{S}$ and those of $\mathcal{L}$. For from

$$f_1(x, y, \lambda_1 x + \lambda_2 y) = g(\lambda; xy), \quad f_2(x, y, \lambda_1 x + \lambda_2 y) = g(\lambda; xy)$$

it follows necessarily that

$$f_1(xyz) = f_2(xyz).$$

Furthermore, every relation in $\mathcal{S}$ is preserved between the uniquely corresponding forms in $\mathcal{L}$ (and conversely). For from

$$c_1 g^{(1)}(\lambda; xy) + c_2 g^{(2)}(\lambda; xy) + \cdots + c_r g^{(r)}(\lambda; xy) = 0$$

it follows by the lemma that necessarily

$$c_1 f^{(1)}(x, y, z) + c_2 f^{(2)}(x, y, z) + \cdots + c_r f^{(r)}(x, y, z) = 0.$$

This proves equality of the ranks of $\mathcal{L}$ and $\mathcal{S}$; $\rho = \sigma$.

b) For comparing the ranks of $\mathcal{S}$ and $\mathcal{T}$ we use the corresponding Lemma b): from $h(x, y, z) = 0$ it follows necessarily that $g(\lambda; xy) = 0$. For the proof, observe that by 3. $h(x, y, z) = 0$ is equivalent to the vanishing of the individual power products

$$\left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_1} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_1} \right)^{p_1} \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_2} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_2} \right)^{p_2} g(\lambda; x, y), \quad p_1 + p_2 = p.$$

¹⁰As in I,

$$\left(t \frac{\partial}{\partial x} \right) = t_1 \frac{\partial}{\partial x_1} + t_2 \frac{\partial}{\partial x_2},$$

and hence in particular

$$\begin{aligned} \left[(\lambda_1 x + \lambda_2 y) \frac{\partial}{\partial z} \right] &= (\lambda_1 x_1 + \lambda_2 y_1) \frac{\partial}{\partial z_1} + (\lambda_1 x_2 + \lambda_2 y_2) \frac{\partial}{\partial z_2}, \\ \left[z \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y} \right) \right] &= z_1 \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_1} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_1} \right) + z_2 \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_2} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_2} \right). \end{aligned}$$

Further differentiation then also yields the vanishing of the power products

$$\begin{aligned} & \left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \left(\frac{\partial}{\partial x_2} \right)^{\alpha_2} \left(\frac{\partial}{\partial y_1} \right)^{\beta_1} \left(\frac{\partial}{\partial y_2} \right)^{\beta_2} \\ & \cdot \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_1} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_1} \right)^{p_1} \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x_2} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y_2} \right)^{p_2} g(\lambda; xy). \end{aligned}$$

Consequently every linear combination of these power products vanishes, and hence every form

$$f \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial x} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial y} \right) g(\lambda, xy).$$

Choosing f in particular so that

$$f(x, y, \lambda_1 x + \lambda_2 y) = g(\lambda; xy),$$

one also obtains

$$g \left(\frac{\partial}{\partial x}; \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) g(\lambda, x, y) = 0;$$

or, if

$$g(\lambda, xy) = \sum G_i \lambda_1^{\nu_1} \lambda_2^{\nu_2} x_1^{\alpha_1} x_2^{\alpha_2} y_1^{\beta_1} y_2^{\beta_2},$$

where the G_i are rational-integer linear forms in the indeterminates a , then

$$\sum \nu_1! \nu_2! \alpha_1! \alpha_2! \beta_1! \beta_2! G_i^2 = 0,$$

and hence $g(\lambda; xy) = 0$.

It follows from Lemma b), exactly as in a), that *the ranks of $\mathcal{S}$ and $\mathcal{T}$ are equal; $\sigma = \tau$.*

Thus by a) and b) one has $\rho = \tau$, and consequently — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — the equivalence of these two linear families is proved. Hence every form of $\mathcal{L}$, in particular also θ , can be represented as a linear rational-integer combination of ρ linearly independent forms $h(x, y, z)$:

$$\theta = \sum c_i h^{(i)}(x, y, z) = \sum PZ.$$

This identity, derived for indeterminate coefficients a of θ , remains valid when the indeterminates a are replaced by quantities of any domain of rationality; hence *the reduction theorem is generally proved for the case of three binary rows.*

The general proof, for N rows of n variables each, is completely parallel. Let $\theta(A)$ have dimension α_i in the row A_i . We again consider the three linear families of forms:

1. the family $\mathcal{L}$, consisting of all forms $f(A)$ which arise from $\theta(A)$ by polar processes P and, in each individual row A_i , have the same dimension as $\theta(A)$;
2. the family $\mathcal{S}$, consisting of all forms $g(\lambda; A_1 \cdots A_n)$ which arise from $f(A)$ by the substitutions

$$\begin{aligned} A_{n+1} &= \lambda_{n+1}^{(1)} A_1 + \lambda_{n+1}^{(2)} A_2 + \cdots + \lambda_{n+1}^{(n)} A_n, \\ &\vdots \\ A_N &= \lambda_N^{(1)} A_1 + \lambda_N^{(2)} A_2 + \cdots + \lambda_N^{(n)} A_n; \end{aligned}$$

in this process, the coefficients of the individual power products of the λ 's in $g(\lambda; A_1 \cdots A_n)$ give forms Z of identity (1);

3. the family $\mathcal{T}$, consisting of all forms $h(A)$ defined by

$$\begin{aligned} h(A) &= \left[A_{n+1} \left(\frac{\partial}{\partial \lambda_{n+1}^{(1)}} \frac{\partial}{\partial A_1} + \cdots + \frac{\partial}{\partial \lambda_{n+1}^{(n)}} \frac{\partial}{\partial A_n} \right) \right]^{\alpha_{n+1}} \\ &\cdots \left[A_N \left(\frac{\partial}{\partial \lambda_N^{(1)}} \frac{\partial}{\partial A_1} + \cdots + \frac{\partial}{\partial \lambda_N^{(n)}} \frac{\partial}{\partial A_n} \right) \right]^{\alpha_N} g(\lambda; A_1 \cdots A_n). \end{aligned}$$

Here $h(A)$ is a sum of forms PZ .

Because of the identical relation between any $n + 1$ rows, Lemma a) remains valid; Lemma b) likewise remains valid, since the number of variables did not enter into its proof. Thus $\mathcal{L}$ and $\mathcal{T}$ have the same rank and — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — are identical. Consequently the identity

$$\theta = \sum PZ$$

holds for indeterminate coefficients, and hence for coefficients that are quantities of any domain of rationality; *the reduction theorem is thereby proved in general.*

III.

We still indicate briefly how analogous considerations also yield the *general series expansion* (for $N \leq n$). Let Z be a separately homogeneous form in the n rows $A_1, \dots, A_n$ of n quantities each; and let Δ be the determinant of these rows. We first prove the identity corresponding to (1):

$$Z = \sum PH \pmod{\Delta}, \quad (2)$$

where the forms H contain only the rows $A_1, \dots, A_{n-1}$ and are derived from Z by processes P .

The proof consists in converting the relations derived in II into congruences modulo Δ . For simplicity take three ternary rows x, y, z ; the coefficients are assumed to be indeterminates.

The three linear families $\mathcal{L}, \mathcal{S}, \mathcal{T}$ are defined as in II for the case of binary rows; let their ranks be ρ, σ, τ , while ρ^* and τ^* denote the ranks of $\mathcal{L}$ and $\mathcal{T}$ modulo Δ (that is, there are ρ^* , but not $\rho^* + 1$, forms of $\mathcal{L}$ that are linearly independent mod Δ). Then, as in II, one has

Lemma a): from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ it follows necessarily that $f(xyz) \equiv 0 \pmod{\Delta}$ (and conversely). For, by the relation between four ternary rows,

$$\Delta_1 x - \Delta_2 y + \Delta_3 z = \Delta t, \quad \Delta_1 = (yzt), \dots,$$

there is always a linear relation modulo Δ between three ternary rows:

$$\Delta_1 x - \Delta_2 y + \Delta_3 z \equiv 0 \pmod{\Delta},$$

and hence, as in II,

$$f(x, y, -\Delta_1 x + \Delta_2 y) = \Delta_3^p f(x, y, z) \pmod{\Delta}.$$

It follows from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ (identically in λ), since Δ is irreducible and Δ_3 is not divisible by Δ , that necessarily $f(x, y, z) \equiv 0 \pmod{\Delta}$. The converse is clear from $\Delta(x, y, \lambda_1 x + \lambda_2 y) = 0$. Hence, as in II, $\rho^* = \sigma$.

Lemma b) remains valid with its proof, and one has therefore $\sigma = \tau$.

For proving $\tau = \tau^*$ we use

Lemma c): from $h(x, y, z) \equiv 0 \pmod{\Delta}$ it follows necessarily that $h(x, y, z) = 0$. Namely, as a polar, $h(x, y, z)$ vanishes under the application of the known Ω -process; this is expressed by the differential equation

$$\Delta \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) h(x, y, z) = 0.$$

If now $h(x, y, z) = \Delta(x, y, z) \cdot k(x, y, z)$, then by further differentiation one also has the vanishing of

$$k \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \Delta \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = h \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) h(x, y, z),$$

and hence of $h(x, y, z)$. Thus $\rho^* = \sigma = \tau = \tau^*$; since $\mathcal{T}$ is a subfamily of $\mathcal{L}$, identity (2) is proved. The same argument gives the proof for n variables.¹¹

Applying identity (2) to the multiplier of Δ , one obtains an expansion

$$Z = \varphi_0 + \Delta \varphi_1 + \Delta^2 \varphi_2 + \dots + \Delta^\mu \varphi_\mu,$$

¹¹After 3. one has the corresponding relation, vanishing because of the determinant factor, for $\Omega(h)$; in operator notation it is the equation arising from products of

$$\left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta} \right), \quad \left[z \left(\frac{\partial}{\partial \lambda_1} \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \lambda_2} \frac{\partial}{\partial \eta} \right) \right],$$

and it vanishes because of the vanishing of the determinant factor.

where the φ_i each vanish under application of the Ω -process. The i -fold repetition of the process gives, since in general $\Omega(\Delta \cdot \psi) = c_1 \psi + c_2 \Delta \cdot \Omega(\psi)$,

$$c \cdot \Omega^i(Z) \equiv \varphi_i \pmod{\Delta};$$

on the other hand, by (2) one has

$$c \cdot \Omega^i(Z) \equiv \sum PH_i \pmod{\Delta},$$

where the H_i are derived from the form $\Omega^i(Z)$ in the same way as H is derived from Z . By Lemma c) (extended to n variables) one therefore gets $\varphi_i = \sum PH_i$, and consequently

$$Z = \sum PH + \Delta \cdot \sum PH_1 + \Delta^2 \cdot \sum PH_2 + \cdots + \Delta^\mu \cdot \sum PH_\mu \quad (3)$$

as the most general series expansion of a form in n rows of n variables each, according to polars of forms in only $n - 1$ rows.

The series expansion for forms in N rows of n variables ($N < n$) is then obtained, as is known, by a simple substitution. We put (for $N = 3$, $n > 3$)

$$u = \xi_1 x + \xi_2 y + \xi_3 z, \quad v = \eta_1 x + \eta_2 y + \eta_3 z, \quad w = \zeta_1 x + \zeta_2 y + \zeta_3 z,$$

and expand $H(u, v, w) = Z(\xi, \eta, \zeta)$ according to (3). Then, since ξ, η, ζ occur only in the combinations u, v, w , one has

$$\left(\eta \frac{\partial}{\partial \xi} \right) = \left(v \frac{\partial}{\partial u} \right) \cdots, \quad \Omega = \sum_{ikl} \left(\frac{\partial}{\partial u} \frac{\partial}{\partial v} \frac{\partial}{\partial w} \right)_{ikl} (xyz)_{ikl} = \nabla_{uvw}(xyz).$$

Under the specialization $\xi_1 = \eta_2 = \zeta_3 = 1$; $\xi_2 = \xi_3 = \cdots = \zeta_2 = 0$, $H(u, v, w)$ goes over into $H(x, y, z)$, and $\nabla_{uvw}(xyz)$ into $\nabla_{xyz}(xyz) = \nabla_{xyz}$ up to a numerical factor, since the application of $(y\partial/\partial x)$ etc. to the determinants $(xyz)_{ikl}$ makes them vanish. Since the corresponding assertion holds for N rows, (3) gives the expansion

$$H = \sum P\Phi + \sum P\Phi_1 + \cdots + \sum P\Phi_\mu, \quad (4)$$

where the Φ_i arise from $\nabla_{A_1 \dots A_N}^i H$ by processes P and contain only the rows $A_1, \dots, A_{N-1}$, apart from the determinant combinations occurring in ∇^i , which, as mentioned, do not come into consideration under polarization.

If these determinants in ∇^i are then replaced by indeterminates p , the resulting forms can again be expanded according to (4); in this way one finally obtains an expansion

$$H = \left[\sum P\psi(p) \right]_{p=\text{determinants of } A_1 \dots A_N}, \quad (5)$$

where the ψ arise from the original form by the processes P and $\nabla_{A_1 \dots A_N}(p)$, $\nabla_{A_1 \dots A_{N-1}}(p)$, and so on. With these expansions and their combinations — for example by applying (3), and then (4) and (5), to the forms Z occurring in (1) — all known expansions for forms in arbitrarily many rows of n cogredient variables each are exhausted.

Erlangen, 5 January 1915.